

José R. Sousa

Charlotte, NC 28217

LinkedIn: <https://www.linkedin.com/in/jrsousa2/>

Phone: 904-290-2890

E-mail: jrsousa2@gmail.com

GitHub: <https://jrsousa2.github.io/>

SUMMARY OF QUALIFICATIONS

- Large experience in **database / data warehouse** and **SQL** (query language).
- Great with **Python** programming in **VS Code**. Created executable from code, made **web API** requests to retrieve data, created **GUI's**, developed **REST API's**, loaded **Excel** and **XML** files into **pandas** dataframes, read and wrote to Excel files with **openpyxl**, created formatted **PowerPoint** slides automatically, created **virtual environments**, automated **batch jobs**, etc. [All can be viewed on my GitHub](#).
- Extracted data from **Hadoop** with Dremio / Hive and created VDS's.
- Fitted **logistic regression** models using both **SAS** and **PySpark ML**.
- Independently created **logistic regression** model in **SAS** to score risk of auto policyholders. The model was deployed.
- Created project on fitting a logistic regression with **PySpark** to a public credit risk dataset. That involves ETL to ingest, clean, transform, and prepare the data before modeling.
- Worked on **insurance pricing models** in partnership with consultancies.
- **Expert at SAS** (**Enterprise Guide** and **Viya**) -- created **stored processes** and **ETL's**, wrote advanced code to produce tables and reports, and truly mastered **SAS macros**. Created dashboard with SAS/GRAPH and reports with **Proc Report**.
- Strong with **Machine Learning** models and **data analytics** and familiar with **actuarial analyses** and **actuarial reports**.
- Expert at **MS Excel**, created advanced spreadsheets with **VBA macros**, **automated charts** and **graphs**.
- Created **workflows** in **n8n** for Gen AI.
- Built a workflow in n8n that extracts content, transforms it with Python scripts (JSON/HTML processing, metadata generation, etc.), and loads it into downstream publishing steps. Worked on n8n workflow JSON and HTML documentation, where Python manipulated structured data before producing the final artifacts.
- Built an alert code in Python and scheduled it on GitHub (**serverless** process).
- Most of my experience is in the **insurance** industry (15+ years), P&C, personal and commercial. Created a wide variety of **loss triangles**, estimated **ultimate claims** from **LDF's** (**reserving analysis**), calculated ceded losses from **reinsurance** contracts, calculated earned premiums, created **loss ratio reports** and **in-force policies**, capped **GC** losses, performed **reconciliation** with the GL, etc.
- Exposure to **accounting principles** – using the **GL** Oracle database, created **YTD summary reports** with array formulas in **Excel** to support quarterly filings (10-Q and 10-K) and performed account and profit center level reconciliations with it.
- Graduated in **Math / Statistics** from the **best university in Brazil** (**U of São Paulo**).
- Very analytical, self-taught, can **work independently**, created new formulas in math (e.g. <https://arxiv.org/html/1810.07877v7>).
- **U.S. citizen** (had a **green card** through **NatGen** previously).

SOFTWARE

- **Databases:** **Oracle**, **Teradata**, **SQL Server / SSMS** and their **IDE's**.
- **Big Data:** **Snowflake**, **Databricks**, **PySpark**, **Hadoop**, **Dremio**, **Hive**.
- **Programming:** **Python**, **VBA**, **VBScript**, Unix/Windows batch scripting.
- **Cloud:** Working knowledge of **AWS**, **Azure** and **GCP**.
- **SAS:**
 - Platforms: **Unix**, **Linux**, **Windows Server** and **Viya** (Cloud).

- Modules: BASE, STAT, GRAPH, ACCESS, CONNECT.
- Expert knowledge of most used SAS Procs, including **Proc Report**. Advanced **macro language** and **stored processes**.
- **BI / Predictive Modeling**: SAS **Enterprise Guide**, **Power BI**.
- **Code Editors**: **VS Code**, Jupyter Notebook.
- **Other**: **Excel**, **PowerPoint**, SharePoint, **Git**, **RegEx**, **Jira**, **XML**, **JSON**.

PROFESSIONAL EXPERIENCE

Wells Fargo (**banking**)

Senior Data Analyst / Data Scientist (**contract** / **hybrid**)

10/2022 – 12/2025

- **Second project: 2025**
 - Worked on **AML / Financial Crimes** remediations (CIP / BO / CDD / **KYC**).
 - Built AML remediation populations (Trust, NRM4 and transportation assets).
 - Created population identification workbooks and waterfall charts and presented them to senior directors for approval.
 - Wrote complex **optimized SQL** queries to extract data from **Hadoop** using **Dremio** and **Hive** and from various databases (**Teradata**, **Oracle**, SQL Server, DB2, etc.). Was a go-to person for **challenging** technical questions.
 - Developed **SAS** and **SQL** code to create ad-hoc, time-sensitive data extracts and **reports**.
 - Created code test scripts and worked with code testers to validate the code.
 - **Recognized with a full-time offer at project completion.**
- **First project: 2022 through 2025**
 - Worked on **customer remediations**, responsible for the creation of the final artifacts from which refunds and letters are issued to customers as required by **regulators**, following strict **Confluence** procedures.
 - Managed **SAS** code used to create final remediation artifacts, made changes to the code to add **enhancements**, **optimize performance** and minimize the number of **defects** and findings in **peer reviews**. As an example of enhancement, when a new analyst takes over a remediation, all the **artifacts** and **SAS** tables created by the previous analyst are **automatically** saved to a new subfolder as **backup**.
 - Created **ETL**'s to produce new artifacts by mapping fields from the execution data requirements (EDR) to data tables. Contributed to design **artifact layouts**.
 - Used **GitHub** for code version control and sharing and **DMS** as a secure repository for storing and sharing artifacts.

Truist, Charlotte, NC (**banking**)

Data Analytics SQL/Python (**contract** / **remote**)

09/2021 – 03/2022

- Created rules based **decision tool** for the compliance and data analysts to choose their optimal analytics application as a web form created in **Python**.
- Created and maintained current report on user accesses for the various **proprietary applications** and data repositories (**AD**, **ACL**), for **compliance**.
- Ran reconciliation between **ServiceNow/RITM** and data pulled from the **mainframe** to assess correctness of an automatic ticket creation process for abends (abnormal job termination), and presented findings with **PowerPoint**.
- Ran **SQL** queries in **Snowflake** to extract data from the **data lake/data warehouse**.
- Created reports using the **SAS grid** and produced **documentation** following internal **templates**.

Florida Blue, Jacksonville, FL (**healthcare**)

SAS/Python Developer (**contract** / **remote**)

04/2020 – 08/2021

- Developed **ETL** code to extract data from **Oracle** and **DB2** and others, using **SAS 9.4** on **AIX** (Unix) and **Enterprise Guide**.
- Extracted data from multiple sources to create agent **fraud detection** analytics for the **SIU** dept with various metrics. Created **macro to automate** the formatting of the report for each metric using **Proc Report**.
- Automated and scheduled various **SAS ETL** jobs in **Unix**, using a **SAS macro**.
- Used **HTML** syntax in **SAS** to compose rich text e-mail alerts.
- Combined Windows **batch**, **WinScp** and **VB** scripts to schedule jobs and automate file transfers and moving data between platforms using SFTP.
- Created a neat **SAS stored process** for various canned reports.
- Created **visualizations** and **reports** using **Power BI** and **Python**, using pre-modeled datasets (no need for DAX customization).
- Gave **training** to team members on how to design **SAS stored processes**, covering how to create the **GUI** and how to handle the variable **user's inputs**.

National General (Allstate), New York, NY (insurance)
Insurance Data Programmer / Data Manager

04/2015 – 12/2019

- **Reported to the holding company's chief actuary.**
- Mastered the **claims data system** of the company (EPIC) and legacy data stored in **Oracle** and **SQL Server** databases.
- **Planned** and **created** datasets in **SAS** by extracting and combining data from various databases, from which loss triangles and **data analytics** were produced.
- Created **actuarial loss triangles** used in reserving analyses with **SAS macros**, and **reconciled** the numbers with the company's **general ledger**.
- Created quarterly company-wide **YTD summary report** in **Excel** using **general ledger data** extracted with **Oracle BI Discoverer** to support quarterly filings and the annual report preparation (**10K**, **10Q**), for the **SEC** and other **regulators**.
- Participated in the **ATLAS project**, with a goal to capture claim and policy level data into the **GL Oracle database**.
- Extracted data from the **SAS server** to create reports with **SAS macros** and **code**. These reports were used internally or provided to consultants, **auditors** and **regulators**.
- Using **SAS**, ran **statistical data analyses** to track performance of products.
- Created complete documentation about work done on legacy data from acquired companies.

AIG, New York, NY (insurance)
Data Scientist / SAS Programmer (contractor / FTE)

12/2011 – 12/2014

- Planned and extracted data from multiple sources (**mainframe**, **DB2**, **SQL Server** and **Oracle**) to produce **enhanced tables** used in all **reports** and **models** and **processes**, using **ETL** and **SAS/Access** interface to **OLE DB** and **ODBC**.
- Planned the layout and created enhanced tables from which loss triangles, actuarial reports, predictive models (**AQI**) and other processes were fed.
- Created **ETL's** to read **VB/VSAM** files on **IBM mainframe** and load them into **SAS** tables on Windows. These files became the data foundation of the **Global Actuarial Pricing** department.
- **Reconciled** quarterly actuarial data loads created from ETL's with control totals.
- Wrote **SAS macro** to create a wide variety of **loss triangles** used to derive **LDF's** and **estimate** the **ultimate claims**.
- Worked in the **deep dive** and **profitability analyses** of the book of business, by extracting data on **losses**, **premium**, **policy** and **insured attributes** to generate various reports and metrics and to monitor the **performance** of the various **lines**

- of business and segments. Performed claim analytics and policy analytics.
- Created smart algorithm for linking policies based on insured and policy characteristics in SAS. This policy linking was the basis for the creation of a scoring model called AQI (Account Quality Index).
- Worked with Deloitte consultants and provided them with loss triangles used as inputs into the Commercial Auto pricing tool. Also worked with Deloitte on the Specialty Worker's Comp model, making sure that deliverables such as SAS code, data and results of the predictive model met the requirements.
- Created and scheduled weekly jobs to extract submission data entered by underwriters into the AQI model and update a Qlikview dashboard.

Capital One, Toronto, ON (banking)

- Data Analyst (contractor)

01/2011 – 11/2010

- Worked on various AML efforts (KYC / CDD), using Excel/VBA and PowerPoint as main reporting tools.
- Created advanced dashboard with SAS/Graph which ran weekly on Unix and was sent to a distribution list to monitor web performance.
- Tweaked and ran some regular critical processes of the company, such as, file transfers to vendors and others, via Unix scripting (ksh, sh, SQL).
- Created ETL jobs and scheduled them to run in Unix to load daily files received from vendors into SAS and Teradata.
- Created Excel workbook with VBA automation to retrieve data from tables in Teradata and copy formula results from this workbook into text boxes and graphs in a PowerPoint deck used in the AML monitoring process.
- Automated customer bookings report with SAS Proc Report.

Aviva Canada, Toronto, ON (insurance)

- Senior Data Analyst (contractor / FTE)

05/2008 – 12/2010

- Independently created logic and SAS code to extract in-force policies from the commercial lines book of business.
- Independently designed SAS stored processes to automate the creation of loss ratio reports with a straightforward user-friendly GUI.
- Planned and provided data used by EEA consultants to develop risk and pricing models, with accurate match between policies and claims.
- Planned, wrote and ran SAS code used to reconcile monthly actuarial data loads.
- Created extensive documentation on all existing data and processes via HTML and VBA, such as table layouts, variables description and distribution, etc.
- Investigated and troubleshooted a lot of data problems reported by the users.
- Created facts and dimensions tables with SAS OLAP Cube Studio.
- Tweaked and updated monthly actuarial reports with BusinessObjects.
- With support from the actuaries, designed and coded the Fleet Rate Tracking Report in SAS, a project that required rates to be deconstructed with the use of VBA coding.

EDUCATION

Masters's in Statistics - University of São Paulo, Brazil

Bachelor's in Statistics - University of São Paulo, Brazil

Bachelor's in Mathematics - University of São Paulo, Brazil